Start Early & Compounding

LESSON EXPLAINER · BEGINNER · GROWTH CAPITAL ACADEMY

What Is Compounding?

Compounding just means your money makes money — and then **that** money starts making money too. Imagine you invest $100 and it grows by 10% in a year. You now have $110. The next year, that 10% growth applies to the full $110, not just your original $100 — so you earn $11 instead of $10. It sounds small, but stretched over decades, that snowball effect becomes enormous.

Think of a snowball rolling down a long hill. At the top, it barely grows — there isn't much snow to stick to it yet. But the further it rolls, the bigger it gets, and the bigger it gets, the *faster* it picks up even more snow. Money invested early is that snowball right at the top of a very long hill.

Real Example: Sarah vs. Mike

Sarah starts investing $200 a month at age 25. Mike waits ten years and starts investing $400 a month — twice as much — at age 35. Both keep going until they retire at 65, and both earn a typical long-run average return of 7% a year.

	Sarah — starts at 25	Mike — starts at 35
Monthly investment	$200	$400
Years invested	40 years	30 years
Total contributed	$96,000	$144,000
Estimated value at 65	$524,963	$487,988

Even though Mike put in **50% more of his own money**, Sarah still ends up ahead — purely because her money had ten extra years to compound. That's the entire lesson in one example.

The Power of Reinvesting the Gains

A one-time $10,000 investment growing at 7% a year, compared two ways: reinvesting each year's growth (compound) versus taking the same 7% only on your original $10,000 every year (simple). Same rate, same starting amount — the only difference is whether the gains stay invested.

Year	Compound (reinvested)	Simple (not reinvested)	Extra from reinvesting
Year 1	$10,700	$10,700	**+$0**
Year 2	$11,449	$11,400	**+$49**
Year 3	$12,250	$12,100	**+$150**
Year 4	$13,108	$12,800	**+$308**
Year 5	$14,026	$13,500	**+$526**

Year	Compound (reinvested)	Simple (not reinvested)	Extra from reinvesting
Year 6	$15,007	$14,200	**+$807**
Year 7	$16,058	$14,900	**+$1,158**
Year 8	$17,182	$15,600	**+$1,582**
Year 9	$18,385	$16,300	**+$2,085**
Year 10	$19,672	$17,000	**+$2,672**
Year 11	$21,049	$17,700	**+$3,349**
Year 12	$22,522	$18,400	**+$4,122**
Year 13	$24,098	$19,100	**+$4,998**
Year 14	$25,785	$19,800	**+$5,985**
Year 15	$27,590	$20,500	**+$7,090**
Year 16	$29,522	$21,200	**+$8,322**
Year 17	$31,588	$21,900	**+$9,688**
Year 18	$33,799	$22,600	**+$11,199**
Year 19	$36,165	$23,300	**+$12,865**
Year 20	$38,697	$24,000	**+$14,697**
Year 21	$41,406	$24,700	**+$16,706**
Year 22	$44,304	$25,400	**+$18,904**
Year 23	$47,405	$26,100	**+$21,305**
Year 24	$50,724	$26,800	**+$23,924**
Year 25	$54,274	$27,500	**+$26,774**
Year 26	$58,074	$28,200	**+$29,874**
Year 27	$62,139	$28,900	**+$33,239**
Year 28	$66,488	$29,600	**+$36,888**
Year 29	$71,143	$30,300	**+$40,843**
Year 30	$76,123	$31,000	**+$45,123**

Based on a $10,000 starting amount at a 7% annual return — the same illustrative figure used above, not a guarantee of future performance.

Worth knowing: *the 7% figure above is a simplified, illustrative long-run average — real markets go up and down year to year, and past performance never guarantees future returns. The point isn't the exact number; it's the principle that time is one of the most powerful tools you have.*

Quiz Answer Key

1. Compounding means your returns start earning their own returns over time.
2. An extra decade gives compounding more time to work — that's why starting early matters so much.
3. Sarah ends up ahead because her contributions had 10 extra years to compound, not because of better investment choices.
4. Delaying investing reduces the time compounding has to work, which can be very hard to fully make up later.
5. Main takeaway: time in the market is one of the most powerful tools for building wealth.